

AI Foundation Models & Target Safety

Core Learnings from Changping Laboratory 25th HGP Symposium

Beijing · August 24–25, 2026

Core Focus: Regulatory Foundation Models, Statistical AI Agents & Target De-risking

Team: Data Science & Computational Biology

1. The Core Shift: From Sequence Reading to Causal AI

Genomics is crossing its third major threshold:

Past: Observational Correlation

- **HGP (2001)**: Static reference genome map.
- **GWAS / Biobanks**: Statistical associations (P-values) without causal mechanism.
- **Safety Triage**: Static literature queries and bulk gene-constraint scores (pLI).

Present: Predictive & Causal AI

- **Single-Cell Kinetics**: Absolute biophysical binding affinities on native DNA.
- **Genomic Foundation Models**: Predicting functional shifts (ΔK_d) in silico.
- **Safety Triage**: Multi-agent causal synthesis gated by statistical falsification.

Core Objective: Upgrade Target Safety Landscape (TSL) from retrospective search to **mechanistic liability prediction**.

2. Regulatory Foundation Models: Decoding Non-Coding Toxicity

- **Biological Reality:** >90% of GWAS and safety-associated variants lie in non-coding regulatory DNA controlling Transcription Factor (TF) binding.
- **AI Solution (Seq2Kd / FOODIE Paradigm — Xiaoliang Xie & Mingchen Chen, CPL):**
 - Measures absolute single-molecule TF-DNA affinities on native human genomes.
 - Trains deep learning foundation models (Seq2Kd) to predict binding affinity changes across ~500 understudied TFs.
- **Target Safety Impact:**
 - **Predicts Regulatory Collapses:** Identifies whether target perturbation strips critical TFs from vital survival genes.
 - **Mechanistic Non-Coding TSL:** Translates obscure regulatory eQTLs into quantified biophysical disruption scores (ΔK_d).

3. Statistically Gated AI Agents: Eliminating Hallucinations

- **Biological Reality:** Generic LLMs hallucinate plausible biological mechanisms, generating false-positive safety flags that derail viable drug targets.
- **AI Solution (Fever GPT Architecture — Xihong Lin, Harvard):**
 - Maps 9 billion functional variant annotations across 3 billion genomic loci.
 - Couples generative LLM agents with **automated statistical falsification gates**.
- **Target Safety Impact:**
 - **Verifiable Evidence Synthesis:** Every safety claim extracted by the LLM is programmatically validated against cis-pQTL, PheWAS, and gnomAD constraint matrices.
 - **Zero-Hallucination Audit Trail:** Distinguishes true causal safety redlines from random linkage-disequilibrium noise.

4. Somatic Evolution: Separating Adaptation from Cancer

- **Biological Reality:** Detecting clonal growth in target tissue often triggers automatic carcinogenicity red flags, terminating safe programs prematurely.
- **AI & Evolutionary Solution (Peter Campbell, Quotient / ex-Sanger):**
 - Normal aging tissue accumulates 30–40 mutations/cell/year; clonal mosaicism is ubiquitous past age 70.
 - **Key Finding:** Chronic stress-driven mutations (e.g. *FOXO1*, *GPAM* in fatty liver) preserve tissue function and are **counter-selected in cancer**.
- **Target Safety Impact:**
 - **Somatic Fitness Mapping:** Classifies whether target-driven clonal expansion represents benign physiological remodeling or malignant transformation.
 - **De-risks Chronic Targets:** Prevents premature abandonment of high-value metabolic targets.

5. Next-Gen Target Safety Pipeline: Direct Upgrades

Safety Assessment Layer	Current Standard	AI-Powered Upgrade (Post-CPL 2026)	Pipeline Decision Value
Non-Coding Variants	Bulk gene constraint (pLI, LOEUF)	Quantitative TF binding kinetics (Seq2Kd)	Pinpoints exact off-target regulatory network collapse.
Safety Literature & Agent	Uncalibrated LLM prompts / PubMed	Multi-agent pipelines with statistical gating	Eliminates false-positive toxicity flags with causal proof.
Carcinogenicity & Clones	Blanket stop-flag on clonal growth	Somatic fitness & cancer negative selection	De-risks benign adaptive tissue remodeling in metabolic programs.
Cellular Phenotypes	Isolated single-readout assays	Foundation models mapping sequence to morphology	Links HCS image phenotypes directly to underlying TF rewiring.

6. Strategic Takeaways for Data Science

Three Core Principles for Target Safety

- **1. Quantitative Kinetics > P-Values:** Integrate biophysical foundation models (predicting ΔK_d) to resolve non-coding target liabilities.
- **2. Statistical Falsification > Plausible Narratives:** Bound agentic LLMs with automated genetic cross-checks (cis-pQTL / PheWAS / constraint).
- **3. Evolutionary Grounding > Conservative Fear:** Distinguish benign adaptive tissue regeneration from true oncogenesis using human clonal fitness landscapes.

Bottom Line: The future of target safety lies in replacing qualitative warnings with **statistically gated, biophysically grounded AI models**.